

Hybrid Stacking Models for Corporate Bankruptcy Prediction Using Financial Ratios

Vikrant Mishra, Raj Shakya, Kanwaljot Singh, Anjani Kumar*

Abstract—Accurate prediction of corporate bankruptcy is a key problem in financial risk management and early warning systems. The Polish Companies Bankruptcy dataset from the Emerging Markets Information Service (EMIS) provides 64 financial ratios measured one to five years before bankruptcy, but exhibits severe class imbalance and complex nonlinear relationships between predictors and outcomes. This study investigates a spectrum of models, including classical baselines, deep autoencoder-based models, and hybrid stacking ensembles that combine gradient boosting and neural networks.

The main proposed architecture is a hybrid stacking model in which an XGBoost classifier and a feed-forward neural network act as base learners, and a logistic regression model acts as the meta-learner. Synthetic Minority Oversampling Technique (SMOTE) is used to mitigate class imbalance, and Random Forest-based feature selection and stacked autoencoders (SAE) are explored for dimensionality reduction. Experiments compare models trained on all 64 features with those trained on a reduced feature set. The stacking model trained on all features achieves an accuracy of 99%, ROC AUC of 0.9957, and strong minority-class precision–recall behaviour, while a feature-selected and SAE-based variant attains slightly lower performance with improved efficiency. The results indicate that hybrid stacking with appropriate imbalance handling is a robust framework for bankruptcy prediction.

Index Terms—Bankruptcy prediction, financial distress, imbalanced data, stacking ensemble, XGBoost, neural networks, autoencoder, SMOTE.

I. INTRODUCTION

Corporate bankruptcy can generate substantial losses for creditors, investors, employees, and regulators. Early identification of financially distressed firms is therefore an important objective in credit risk management and prudential supervision. Classical models such as Altman’s Z-score and subsequent variants use linear combinations of a small number of financial ratios to discriminate between bankrupt and non-bankrupt firms [1], [2]. Although these approaches remain influential in practice, they often struggle to capture nonlinear interactions and high-dimensional patterns present in contemporary datasets.

Machine learning (ML) and deep learning (DL) techniques have been increasingly adopted for bankruptcy and financial distress prediction, offering improved flexibility and predictive accuracy [3], [4], [5], [6]. However, several challenges remain. First, most real-world datasets are highly imbalanced, with only a small fraction of firms becoming bankrupt [7], [8],

[9]. Second, financial ratios are often noisy and redundant, raising the risk of overfitting if dimensionality reduction is not handled carefully [10], [11]. Third, model outputs must be reliable in terms of minority-class detection, not only overall accuracy [12], [13].

The present study focuses on the Polish Companies Bankruptcy dataset collected from EMIS and made available via the UCI repository [14], [15]. The dataset contains 64 financial ratios for companies tracked between 2000 and 2013, with multiple prediction horizons from one to five years prior to bankruptcy. The work develops and evaluates a range of models, with particular emphasis on hybrid stacking architectures that combine tree-based ensembles and neural networks.

The main contributions are as follows:

- Construction of a complete modelling pipeline for the Polish Companies Bankruptcy dataset, including cleaning, normalisation, SMOTE-based balancing, and Random Forest-based feature selection.
- Design of two hybrid stacking models: (i) a full-feature stacking model using all 64 ratios, SMOTE, XGBoost, neural networks, and logistic regression; and (ii) a feature-selected and SAE-based stacking model that operates on a compressed 32-dimensional latent representation.
- Detailed comparison with classical baselines and autoencoder-based models, with analysis of confusion matrices, ROC and precision–recall curves, and performance trade-offs between using all features and a reduced subset.

II. LITERATURE REVIEW

A substantial body of research has examined bankruptcy and financial distress prediction using statistical and ML-based approaches. Early work by Altman [1] and Ohlson [2] introduced discriminant analysis and logistic regression frameworks based on a small number of financial ratios. These models remain benchmarks in the literature and practice.

Recent surveys highlight a strong shift toward ML and DL methods. Dasilas and Rigani provide a systematic review of ML techniques in bankruptcy prediction, emphasising ensemble learning and feature engineering [3]. Quinteros-Navarro and Rodríguez survey ML-based bankruptcy prediction models and identify data imbalance and feature redundancy as key challenges [16]. Lombardo *et al.* compile a benchmark dataset and compare a variety of ML algorithms on US-listed firms, emphasising the importance of consistent evaluation protocols [4]. Soukal and colleagues discuss state-of-the-art models and

V. Mishra, R. Shakya, K. Singh, and A. Kumar are with the Cluster Innovation Centre, University of Delhi, Delhi, India.

*Corresponding author: Dr. Anjani Kumar (email: Anjaniverma29@gmail.com).

their limitations in terms of interpretability and robustness [17].

A variety of ML architectures have been proposed for financial distress prediction. Random Forests and gradient boosting models such as XGBoost are frequently used due to their strong performance on structured data [18], [19], [5], [20]. Autoencoders and stacked autoencoders (SAE) have been explored for unsupervised feature learning and dimensionality reduction [10], [11], [21]. Deep recurrent and LSTM-based models have been used to capture sequential or temporal structure in financial statements [22], [23], [24], [25]. Graph neural networks have recently been applied to incorporate inter-firm relationships [26], [27].

Handling class imbalance is a central topic in this domain. SMOTE [8] and its variants are often applied to oversample minority classes [7], [9], [12]. Cost-sensitive learning and ensemble-based imbalance handling have also been investigated [28], [29]. Several works specifically analyse the Polish Companies Bankruptcy dataset [15], [30], [21], making it a useful benchmark for comparing algorithms and imbalance mitigation strategies.

Stacking and hybrid models that combine multiple learners have shown promise in related settings. Soui *et al.* use stacked autoencoders followed by a classifier for bankruptcy prediction [10]. Siswoyo *et al.* propose an ensemble of multilayer perceptrons optimised via random search [31]. Kanász *et al.* combine autoencoders with genetic algorithms to construct ensembles [11]. The present study extends this line of work by integrating SMOTE, feature selection, SAE-based compression, XGBoost, and neural networks in a stacked framework tailored to the Polish dataset.

III. DATASET AND PREPROCESSING

A. Polish Companies Bankruptcy Data

The analysis uses the Polish Companies Bankruptcy dataset from the UCI Machine Learning Repository [14]. The data originate from the EMIS database and cover multiple prediction horizons (1–5 years before bankruptcy) for Polish firms from 2000 to 2013. The dataset provides 64 numerical financial indicators that characterise profitability, leverage, liquidity, efficiency, and capital structure [15]. The indicators cover ratios such as Net Profit/Total Assets, Total Liabilities/Total Assets, Working Capital/Total Assets, liquidity measures, multi-year profitability ratios, and several turnover metrics.

A complete list of all 64 financial indicators, along with their descriptions, is provided in Appendix A (Table VII).

B. Data Cleaning, Normalisation, and Splitting

Raw files for multiple prediction horizons are concatenated into a single dataset. Non-numeric placeholders are converted to missing values and imputed using mean imputation. All features are cast to numeric type for compatibility with downstream models.

To avoid scale dominance, features are normalised using min–max scaling to the range [0, 1]:

$$X' = \frac{X - \min(X)}{\max(X) - \min(X)}. \quad (1)$$

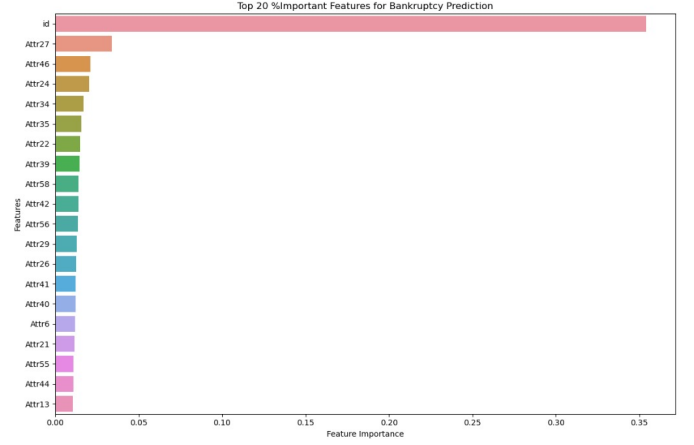


Fig. 1. Top 13 features out of 64 financial ratios based on Random Forest importance.

Feature	Description
Attr22	Profit on Operating Activities / Total Assets.
Attr24	Gross Profit (in 3 Years) / Total Assets.
Attr26	(Net Profit + Depreciation) / Total Liabilities.
Attr27	Profit on Operating Activities / Financial Expenses.
Attr29	Logarithm of Total Assets.
Attr34	Operating Expenses / Total Liabilities.
Attr35	Profit on Sales / Total Assets.
Attr39	Profit on Sales / Sales.
Attr42	Profit on Operating Activities / Sales.
Attr46	(Current Assets - Inventory) / Short-term Liabilities.
Attr56	(Sales - Cost of Products Sold) / Sales.
Attr58	Total Costs / Total Sales.

TABLE I

SELECTED FEATURES AND THEIR DESCRIPTIONS.

The dataset is then split into training (80%) and test (20%) sets using stratified sampling to preserve the original class distribution.

C. Class Imbalance and SMOTE

The dataset is highly imbalanced, with bankrupt firms forming a small minority. Synthetic Minority Oversampling Technique (SMOTE) is applied to the training set to alleviate this imbalance [8]. SMOTE generates synthetic minority samples by interpolating between nearest neighbours in feature space. This procedure is performed only on the training data, leaving the test set unchanged to provide an unbiased evaluation [7], [9].

D. Feature Selection and Autoencoder-Based Reduction

Random Forest feature importance scores are used to rank the 64 ratios and select the most informative subset [18]. The top 13 features are retained, corresponding to approximately the top 20% in importance. Figure 1 shows the ranked importances, and Table I describes the selected attributes.

In addition to feature selection, a stacked autoencoder (SAE) is trained to compress the 64-dimensional input into a 32-dimensional latent representation [25], [10]. The autoencoder minimises the reconstruction loss:

$$L = \frac{1}{n} \sum_{i=1}^n \|X_i - \hat{X}_i\|^2, \quad (2)$$

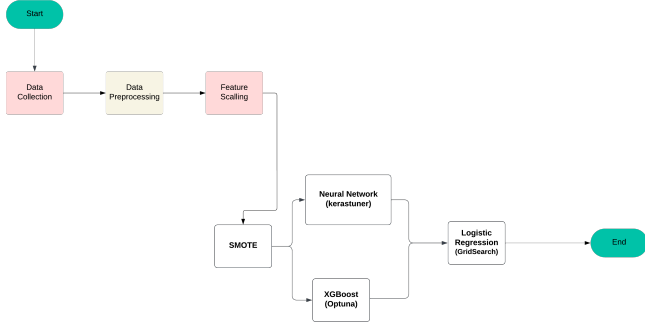


Fig. 2. Workflow of the hybrid stacking model without feature selection (64 features).

where X_i is the original input and $\hat{X}_i$ is the reconstructed output. The encoder maps input X to latent code Z ,

$$Z = \sigma(W_e X + b_e), \quad (3)$$

and the decoder reconstructs $\hat{X}$ from Z ,

$$\hat{X} = \sigma(W_d Z + b_d), \quad (4)$$

with trainable weights W_e, W_d and biases b_e, b_d . The 32-dimensional codes are used as inputs to selected downstream models.

IV. MODELS AND PROPOSED ARCHITECTURES

A. Baseline Models

Several baseline classifiers are trained on scaled features, with and without feature selection:

- Logistic Regression (LR), a linear baseline.
- Random Forest (RF), an ensemble of decision trees [18].
- XGBoost, a gradient boosting ensemble [19].
- Multilayer Perceptron (MLP), a feed-forward neural network with ReLU activations [25].

These baselines provide benchmarks for comparing deep and hybrid models [5], [4].

B. Hybrid Stacking Without Feature Selection

The first proposed architecture uses all 64 features, SMOTE-balanced training data, and a stacking ensemble consisting of an XGBoost classifier and a neural network as base learners, with a logistic regression meta-learner. Figure 2 shows the workflow, and Figure 3 illustrates the architecture.

The neural network comprises an input layer with 64 units, two hidden layers with optimised width, and an output unit with sigmoid activation for binary classification. The pre-activation z and sigmoid output $f(z)$ are

$$z = \sum_{i=1}^n w_i x_i + b, \quad f(z) = \frac{1}{1 + e^{-z}}. \quad (5)$$

Binary cross-entropy loss is used:

$$L = -\frac{1}{m} \sum_{i=1}^m [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]. \quad (6)$$

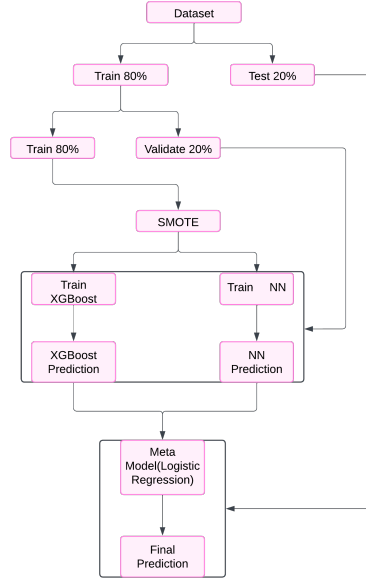


Fig. 3. Proposed hybrid stacking architecture without feature selection (64 features).

The XGBoost component minimises a regularised objective [19]:

$$\mathcal{L}(\Theta) = \sum_{i=1}^m l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k), \quad (7)$$

with tree complexity penalty

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2, \quad (8)$$

where T is the number of leaves and w_j is the weight of leaf j .

The stacking ensemble uses the predicted probabilities from the neural network and XGBoost as meta-features for logistic regression. The meta-learner models the bankruptcy probability as

$$P(y = 1|x) = \frac{1}{1 + e^{-(w^\top x + b)}}. \quad (9)$$

C. Hybrid Stacking With Feature Selection and SAE

A second proposed model combines Random Forest feature selection, SAE-based dimensionality reduction, and stacking. The pipeline first selects 13 important features, then trains an SAE that compresses the 64 original features into a 32-dimensional latent representation. The neural network and XGBoost are trained on the 32 latent features, and their predictions are stacked via logistic regression. Figure 4 depicts the workflow, and Figure 5 shows the corresponding architecture.

The SAE loss and encoding/decoding functions are given in Section III-D. Training is performed using binary cross-entropy with an Adam optimiser [32], [25].

D. Hyperparameter Tuning

Hyperparameters are tuned using a combination of Keras Tuner, Optuna, and GridSearchCV [33]. Table II summarises the tuning strategies.

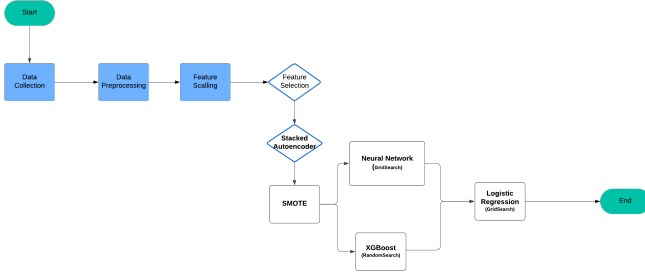


Fig. 4. Workflow of the hybrid stacking model with feature selection and SAE-based dimensionality reduction.

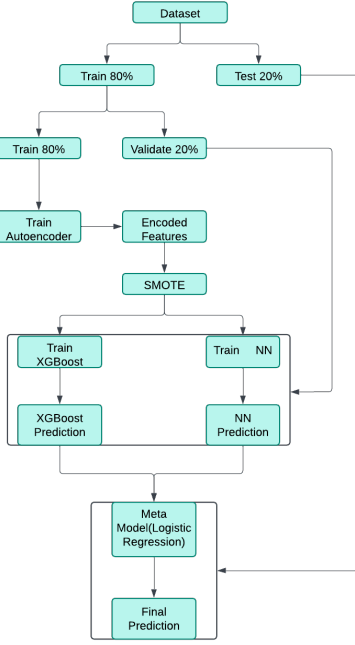


Fig. 5. Proposed hybrid stacking architecture with feature selection and SAE (32 latent features).

Model	Technique	Purpose
Neural Network	Keras Tuner	Search over depth, units, and dropout.
XGBoost	Optuna	Optimise learning rate, depth, estimators.
Logistic Regression	GridSearchCV	Tune regularisation strength.

TABLE II

HYPERPARAMETER TUNING TECHNIQUES AND THEIR APPLICATIONS.

V. EXPERIMENTAL SETUP

All models are implemented in Python using scikit-learn [33], XGBoost [19], TensorFlow/Keras [25], and the imbalanced-learn toolkit for SMOTE [8]. The dataset is split into training and test sets with an 80/20 ratio and stratification by the target label. Scaling and SMOTE are applied only to the training data.

Evaluation metrics include accuracy, precision, recall, F1-score, ROC AUC, and precision-recall AUC [7], [12]. Confusion matrices, ROC curves, and precision-recall curves are used to analyse classification behaviour for both majority and

Model	Acc.	AUC	Prec. (0)	Prec. (1)
Logistic Regression	0.95	0.91	0.95	0.97
Neural Network	0.97	0.97	0.99	0.43
XGBoost	0.98	0.99	0.99	0.89
Random Forest	0.98	0.99	0.98	0.92
SAE + XGBoost + NN + LR	0.97	0.97	0.99	0.62

TABLE III

PERFORMANCE METRICS FOR MODELS WITH FEATURE SELECTION.

Model	Acc.	AUC	Prec. (0)	Prec. (1)
SAE + LSTM	0.9668	0.9700	0.97	0.78
Autoencoder + XGBoost	0.9667	0.9723	0.98	0.75
SMOTE + SAE + XGB + ANN	0.9315	0.9341	0.99	0.40
Hybrid Stacking (XGB + NN + LR)	0.99	0.9957	0.99	0.85

TABLE IV

PERFORMANCE METRICS FOR MODELS WITHOUT FEATURE SELECTION.

minority classes.

VI. RESULTS AND DISCUSSION

A. Models With Feature Selection

Table III reports performance metrics for models trained on the feature-selected dataset (13 features). The hybrid stacking model with SAE and logistic regression is included for comparison.

Tree-based ensembles exhibit strong performance, with XGBoost and Random Forest achieving 98% accuracy and AUC values near 0.99. The hybrid SAE + XGBoost + NN + LR model delivers competitive performance but somewhat lower precision for the minority class.

B. Models Without Feature Selection

Table IV summarises the results for deep and hybrid models trained on all 64 features. The SAE + LSTM and Autoencoder + XGBoost models illustrate the impact of autoencoder-based compression [10], [23].

The hybrid stacking model without feature selection achieves the best performance, combining very high overall accuracy with strong minority-class precision and AUC.

C. Hybrid Stacking Without Feature Selection: Detailed Analysis

Figure 6 shows the confusion matrix for the 64-feature stacking model. True negatives and true positives are high, while the number of false negatives (missed bankruptcies) is relatively low.

Training and validation loss curves in Figure 7 indicate stable convergence and limited overfitting. Accuracy curves in Figure 8 show validation accuracy stabilising near 98%.

The precision-recall curve in Figure 9 shows an average precision of 0.9438, reflecting strong performance on the minority class. The ROC curve in Figure 10 achieves an AUC of 0.9957, indicating excellent discrimination between bankrupt and non-bankrupt firms.

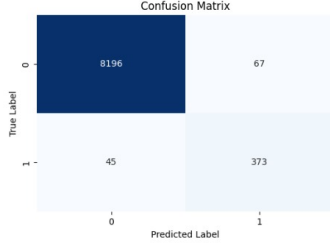


Fig. 6. Confusion matrix for the hybrid stacking model without feature selection.

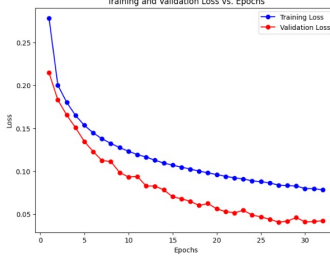


Fig. 7. Training and validation loss vs. epochs for the 64-feature stacking model.

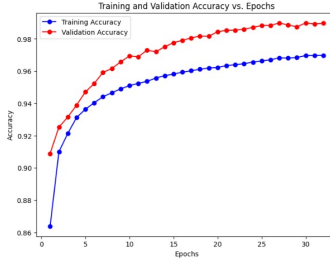


Fig. 8. Training and validation accuracy vs. epochs for the 64-feature stacking model.

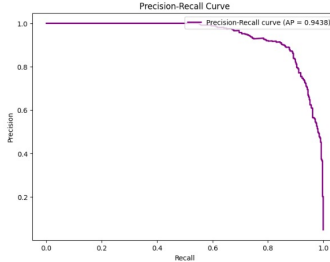


Fig. 9. Precision–recall curve for the 64-feature stacking model (AP = 0.9438).

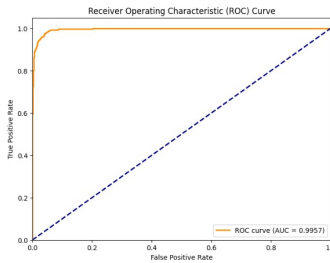


Fig. 10. ROC curve for the 64-feature stacking model (AUC = 0.9957).

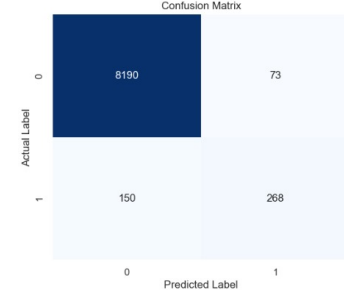


Fig. 11. Confusion matrix for the hybrid stacking model with feature selection and SAE.

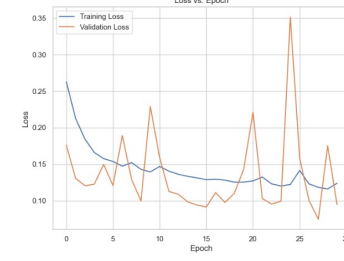


Fig. 12. Training and validation loss vs. epochs for the feature-selected stacking model.

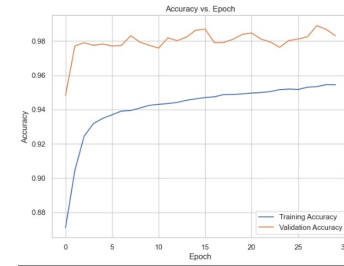


Fig. 13. Training and validation accuracy vs. epochs for the feature-selected stacking model.

D. Hybrid Stacking With Feature Selection and SAE: Detailed Analysis

Figures 11–15 present corresponding plots for the feature-selected and SAE-based stacking model. The confusion matrix in Figure 11 shows higher false negatives compared to the full-feature model, reflecting a loss of sensitivity to bankrupt firms.

Training and validation loss (Figure 12) and accuracy (Figure 13) curves indicate good convergence and generalisation, with validation accuracy around 98%.

The precision–recall curve in Figure 14 has an AUC of 0.8192, lower than the full-feature model but still reflecting reasonable performance. The ROC curve in Figure 15 achieves an AUC of 0.9863.

E. Comparison of Configurations With and Without Feature Selection

Table V compares key metrics for the two stacking configurations. The full-feature model achieves higher true positives, lower false negatives, and better precision–recall AUC,

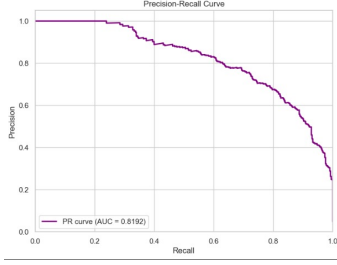


Fig. 14. Precision–recall curve for the feature-selected stacking model (AP = 0.8192).

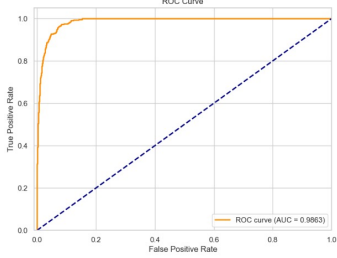


Fig. 15. ROC curve for the feature-selected stacking model (AUC = 0.9863).

Metric	With FS (12 Features)	Without FS (64 Features)
Training Accuracy (%)	96	97
Validation Accuracy (%)	98	98
Training Loss	0.05	0.04
Validation Loss	0.06	0.05
True Positives (TP)	268	373
True Negatives (TN)	8190	8196
False Positives (FP)	73	67
False Negatives (FN)	150	45
Precision–Recall AUC	0.8192	0.9438
ROC AUC	0.9863	0.9957

TABLE V

PERFORMANCE COMPARISON OF STACKING MODELS WITH AND WITHOUT FEATURE SELECTION.

whereas the feature-selected model offers computational savings and a more compact representation.

These results indicate that the 64-feature stacking model is preferable when maximising bankrupt-firm detection is critical. The feature-selected variant remains attractive under resource constraints or when interpretability and compactness are prioritised.

F. Comparison With Existing Literature

Table VI compares the proposed stacking models with representative approaches from the literature, including hybrid XGBoost + ANN and SAE + Softmax architectures [10], [5], [21], [11]. The full-feature stacking model achieves competitive or superior AUC and F1-scores relative to previously reported results on similar datasets.

These findings align with broader reports that stacking ensembles can outperform individual base learners, particularly when imbalance mitigation and representation learning are integrated [28], [29], [12], [34].

Model	Acc. (%)	AUC	F1	Notes
XGBoost + ANN	98.3	0.958	0.89	Hybrid sequential model [5].
SAE + Softmax	97.5	0.890	0.85	SAE-based model [10].
Hybrid Stacking (FS + SMOTE + SAE + XGB + NN + LR)	97.4	0.9863	0.918	Feature-selected stacking.
Hybrid Stacking (SMOTE + XGB + NN + LR, 64 feats)	99.0	0.9957	0.899	Proposed full-feature stacking model.

TABLE VI

COMPARISON OF PROPOSED MODELS WITH REPRESENTATIVE LITERATURE BASELINES.

VII. CONCLUSION

This study has presented a comprehensive pipeline for corporate bankruptcy prediction on the Polish Companies Bankruptcy dataset, integrating data cleaning, scaling, SMOTE-based balancing, feature selection, autoencoder-based dimensionality reduction, and hybrid stacking ensembles. The main proposed model—a stacking ensemble that combines XGBoost and a neural network via a logistic regression meta-learner—achieves an accuracy of 99%, ROC AUC of 0.9957, and strong precision–recall performance for the minority class.

A secondary configuration that combines Random Forest-based feature selection and SAE compression offers competitive performance with reduced dimensionality and computational cost. Comparative analysis against classical baselines and representative literature models confirms the effectiveness of hybrid stacking with appropriate imbalance handling.

Future work can investigate cost-sensitive learning, explainability techniques such as SHAP [30], temporal modelling across multiple years of ratios, and external validation on other national datasets [35], [36], [37]. These directions may further enhance the robustness and applicability of bankruptcy prediction systems in high-stakes financial environments.

APPENDIX A

FINANCIAL INDICATOR DEFINITIONS

TABLE VII: Financial indicators and their descriptions.

Indicator	Description
X1	Net Profit / Total Assets: A measure of profitability, indicating the efficiency of using assets to generate profit.
X2	Total Liabilities / Total Assets: Reflects the company's financial leverage by showing the proportion of assets financed by liabilities.
X3	Working Capital / Total Assets: Indicates the proportion of net working capital to total assets, revealing short-term financial health.
X4	Current Assets / Short-term Liabilities: A liquidity ratio that assesses the ability to cover short-term obligations with current assets.
X5	$[(\text{Cash} + \text{Short-term Securities} + \text{Receivables} - \text{Short-term Liabilities}) / (\text{Operating Expenses} - \text{Depreciation})] * 365$: Measures cash flow adequacy relative to operating expenses over a year.
X6	Retained Earnings / Total Assets: Shows how much profit is retained in the business relative to total assets.
X7	EBIT / Total Assets: Indicates operational profitability relative to the company's total assets.
X8	Book Value of Equity / Total Liabilities: Assesses the company's solvency by comparing equity to liabilities.
X9	Sales / Total Assets: Measures the asset turnover, indicating how efficiently assets are used to generate sales.
X10	Equity / Total Assets: Reflects the proportion of assets financed by equity.
X11	$(\text{Gross Profit} + \text{Extraordinary Items} + \text{Financial Expenses}) / \text{Total Assets}$: Measures overall profitability adjusted for extraordinary items and financial costs.
X12	Gross Profit / Short-term Liabilities: Indicates the ability to cover short-term obligations with gross profit.
X13	$(\text{Gross Profit} + \text{Depreciation}) / \text{Sales}$: Shows gross profit efficiency, including the impact of depreciation.
X14	$(\text{Gross Profit} + \text{Interest}) / \text{Total Assets}$: Measures profitability from gross profit and interest relative to total assets.
X15	$(\text{Total Liabilities} * 365) / (\text{Gross Profit} + \text{Depreciation})$: Calculates the number of days required to cover liabilities from gross profit and depreciation.
X16	$(\text{Gross Profit} + \text{Depreciation}) / \text{Total Liabilities}$: Indicates the company's ability to cover liabilities from gross profit and depreciation.
X17	Total Assets / Total Liabilities: A solvency ratio showing how many times total assets cover total liabilities.
X18	Gross Profit / Total Assets: Assesses the company's operational profitability relative to total assets.
X19	Gross Profit / Sales: Indicates the margin of gross profit relative to sales.
X20	$(\text{Inventory} * 365) / \text{Sales}$: Measures inventory turnover in days.
X21	$\text{Sales}(n) / \text{Sales}(n-1)$: Shows the sales growth rate compared to the previous year.
X22	Profit on Operating Activities / Total Assets: Reflects operational profitability relative to assets.
X23	Net Profit / Sales: A profitability measure indicating the net margin of sales.
X24	Gross Profit (in 3 Years) / Total Assets: Assesses cumulative gross profitability relative to assets over three years.
X25	$(\text{Equity} - \text{Share Capital}) / \text{Total Assets}$: Indicates retained earnings relative to total assets after deducting share capital.
X26	$(\text{Net Profit} + \text{Depreciation}) / \text{Total Liabilities}$: Measures the company's ability to cover liabilities with net profit and depreciation.
X27	Profit on Operating Activities / Financial Expenses: Assesses operational profitability relative to financial costs.
X28	Working Capital / Fixed Assets: Indicates the efficiency of managing working capital relative to fixed assets.
X29	Logarithm of Total Assets: A logarithmic transformation of total assets to reduce data skewness.

Indicator	Description
X30	(Total Liabilities - Cash) / Sales: Measures the proportion of liabilities, excluding cash, relative to sales.
X31	(Gross Profit + Interest) / Sales: Indicates the contribution of gross profit and interest to sales.
X32	(Current Liabilities * 365) / Cost of Products Sold: Assesses the number of days required to cover current liabilities with costs of products sold.
X33	Operating Expenses / Short-term Liabilities: Reflects the ratio of operating expenses to short-term obligations.
X34	Operating Expenses / Total Liabilities: Indicates the operating cost burden relative to total liabilities.
X35	Profit on Sales / Total Assets: Shows profitability from sales relative to total assets.
X36	Total Sales / Total Assets: Measures the efficiency of assets in generating sales.
X37	(Current Assets - Inventories) / Long-term Liabilities: A liquidity ratio indicating the proportion of liquid assets (excluding inventory) to long-term obligations.
X38	Constant Capital / Total Assets: Reflects the stability of capital investment relative to total assets.
X39	Profit on Sales / Sales: Assesses profit margins relative to sales.
X40	(Current Assets - Inventory - Receivables) / Short-term Liabilities: Indicates liquidity after excluding inventory and receivables.
X41	Total Liabilities / ((Profit on Operating Activities + Depreciation) * (12/365)): Measures solvency over a year.
X42	Profit on Operating Activities / Sales: Reflects operational efficiency relative to sales.
X43	Rotation Receivables + Inventory Turnover in Days: Indicates the combined turnover of receivables and inventory in days.
X44	(Receivables * 365) / Sales: Measures receivables turnover in days.
X45	Net Profit / Inventory: Indicates profitability relative to inventory levels.
X46	(Current Assets - Inventory) / Short-term Liabilities: Measures liquidity after excluding inventory.
X47	(Inventory * 365) / Cost of Products Sold: Assesses inventory management efficiency.
X48	EBITDA (Profit on Operating Activities - Depreciation) / Total Assets: Reflects profitability from core operations relative to assets.
X49	EBITDA (Profit on Operating Activities - Depreciation) / Sales: Indicates operational efficiency relative to sales.
X50	Current Assets / Total Liabilities: A liquidity ratio showing the proportion of current assets to total liabilities.
X51	Short-term Liabilities / Total Assets: Reflects short-term financial obligations relative to total assets.
X52	(Short-term Liabilities * 365) / Cost of Products Sold: Measures the time required to cover short-term liabilities with costs of products sold.
X53	Equity / Fixed Assets: Indicates the equity invested in fixed assets.
X54	Constant Capital / Fixed Assets: Reflects the stability of fixed asset financing.
X55	Working Capital: Indicates the net available liquidity.
X56	(Sales - Cost of Products Sold) / Sales: Reflects gross profit margin relative to sales.
X57	(Current Assets - Inventory - Short-term Liabilities) / (Sales - Gross Profit - Depreciation): Measures liquidity efficiency excluding inventory and short-term liabilities.
X58	Total Costs / Total Sales: Indicates the proportion of total costs to sales.
X59	Long-term Liabilities / Equity: Reflects long-term leverage relative to equity.
X60	Sales / Inventory: Indicates inventory turnover.
X61	Sales / Receivables: Reflects receivables turnover efficiency.
X62	(Short-term Liabilities * 365) / Sales: Measures the time required to settle short-term liabilities from sales revenue.
X63	Sales / Short-term Liabilities: Indicates the ability to cover short-term obligations through sales.
X64	Sales / Fixed Assets: Measures the efficiency of fixed assets in generating sales.

APPENDIX B

CODE AND DATA AVAILABILITY

The complete implementation of the bankruptcy prediction pipeline, including data preprocessing, model training, and evaluation notebooks, is available at:

<https://github.com/Vikrantmishra/Bankruptcy-Prediction>

The Polish Companies Bankruptcy dataset used in this study is publicly available from the UCI Machine Learning Repository:

<https://archive.ics.uci.edu/dataset/365/polish+companies+bankruptcy+data>

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